Slipping-contact friction in rattleback dynamics: extension of a rolling formulation using the Moffatt–Shimomura contact laws

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The rattleback is a semi-ellipsoidal solid that spins steadily in one sense but reverses when spun in the other. Formulations that assume rolling without slipping, including the one extended here, eliminate the contact force algebraically and conserve energy. Such models cannot contain friction, because friction depends on the slip velocity of the contact point and that velocity vanishes identically under rolling. Moffatt and Shimomura showed for spinning spheroids that the analogous rise phenomenon is driven by weak friction associated with slipping and is absent for a rolling body. We release the rolling constraint, which raises the governing system from six to nine first-order equations, and construct the contact force explicitly as a normal reaction plus a tangential friction force. The normal reaction is solved from the condition that the body remain on the plane rather than approximated by mg; the result is a closed-form scalar expression whose denominator is bounded below by unity in the frictionless case. In the rolling solution the normal reaction ranges from $0.20\,mg$ to $5.86\,mg$, and holding it at mg shifts the reversal time by 14% near the rolling limit. Three tangential laws are compared: regularized Coulomb friction, viscous friction, and Coulomb friction with a drilling torque. With Coulomb friction at $\mu = 0.15$ the reversal is delayed from 2.83 to 4.22 s and 12% of the energy is dissipated. The Coulomb and viscous laws differ structurally: a viscous law of comparable nominal strength produces no reversal, which mirrors the finite-time and asymptotic transitions of Moffatt and Shimomura. A drilling torque with a Hertzian lever arm leaves the reversal intact and then brings the body nearly to rest, whereas an oversized contact patch suppresses the effect entirely.

I. INTRODUCTION

A rattleback, also called a celt or wobblestone, is a rigid body with a convex base whose principal axes of inertia are rotated slightly, about the vertical, relative to the principal directions of curvature of its surface. This misalignment makes the body chiral and gives it a preferred sense of rotation. Spun in the preferred sense it continues to spin. Spun in the other sense it develops a growing pitching and rolling oscillation, transfers its spin energy into that oscillation, and emerges rotating in the preferred sense.

The phenomenon has a long history. Walker gave the first linear analysis and identified the coupling between the spin and the rocking modes [1]. Bondi's treatment of the misalignment between inertial and geometric axes became the standard account of the one-way stability [2], and averaged equations for the slow evolution of the spin were obtained by Markeev [3] and Pascal [4]. Moffatt and Tokieda presented the reversal as a prototype of chiral dynamics [5]. Nonlinear simulations began with Kane and Levinson, whose parameter set is used here [6], and with Lindberg and Longman [7]. Garcia and Hubbard compared simulations with experiments and included aerodynamic and contact dissipation [8]. Borisov and Mamaev showed that the nonholonomic rolling model can display strange attractors although it conserves energy [9], and Kondo and Nakanishi derived an analytic estimate of the reversal time under the rolling constraint [10].

The formulation extended in this work [11] describes the orientation by a 1–2–3 sequence of Euler angles, assumes rolling without slipping, and integrates six coupled first-order equations. It reproduces the spin reversal, the dependence of the reversal time on the initial spin, and the spin induced by tapping, in agreement with Ref. [6]. Like most rattleback models, however, it imposes rolling. The constraint keeps the equations compact but has two restrictive consequences. The contact point does no work, so energy is conserved and reversals can recur indefinitely, whereas real rattlebacks lose energy at the contact and usually reverse once before coming to rest. And the contact force is eliminated from the equations of motion, so it is never computed and cannot be compared with a friction limit. Dissipative models have been studied with a viscous slip law [12] and with Coulomb–Contensou friction integrated over an elliptic contact patch [13].

The aim of the present work is to admit friction at the contact following the treatment of Moffatt and Shimomura [14]. Their analysis of spinning eggs and spheroids identifies the frictional force between the body and the table as the essential ingredient. If that force vanishes, or if the body rolls without slipping, the rise of the spinning egg does not occur; weak friction associated with slipping at the contact produces a secular instability. The same reasoning applies to the rattleback: a rolling model cannot contain a mechanism driven by slip. Moffatt and Shimomura examined two tangential laws, a Coulomb law and a law linear in the slip velocity, under an approximation in which the normal reaction is of order Mg.

We adopt those two laws for the rattleback, add a drilling torque, and solve for the normal reaction exactly. Section II summarizes the rolling formulation. Section III explains why friction cannot be added to it and derives the nine-state slipping formulation, with the normal reaction in Sec. III C. Section IV describes the implementation and its verification, Sec. V presents the results, and Sec. VI states the limitations.

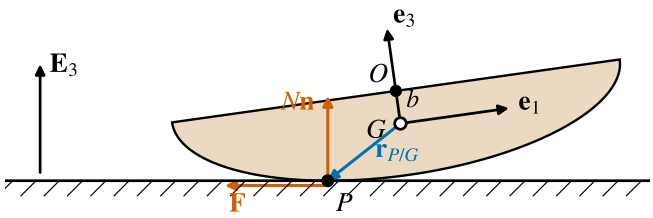


FIG. 1. Section of the rattleback in the e_1–e_3 plane, not to scale and with the tilt exaggerated. O is the center of the ellipsoid, G the center of mass, and P the contact point. In the slipping formulation the contact force consists of a normal reaction $N\boldsymbol{n}$ and a tangential friction force $\boldsymbol{F}$.

II. ROLLING FORMULATION

A. Geometry

Let $\{\boldsymbol{E}_1,\boldsymbol{E}_2,\boldsymbol{E}_3\}$ be a fixed frame with $\boldsymbol{E}_3$ vertical, and let $\{\boldsymbol{e}_1,\boldsymbol{e}_2,\boldsymbol{e}_3\}$ be a body frame with origin at the center of mass G, aligned with the principal directions of curvature (Fig. 1). The base is the lower half of the ellipsoid

$$\frac{x_1^2}{a_1^2}+\frac{x_2^2}{a_2^2}+\frac{(x_3-b)^2}{a_3^2}=1, \tag{1}$$

whose center O lies a distance b above G. With the 1–2–3 Euler angles (ϕ,θ,ψ), the body-frame components of the vertical, $\boldsymbol{n}\equiv\boldsymbol{E}_3=n_i\boldsymbol{e}_i$, are

$$\begin{aligned} n_1&=\sin\phi\sin\psi-\cos\phi\sin\theta\cos\psi,\\ n_2&=\sin\phi\cos\psi+\cos\phi\sin\theta\sin\psi,\\ n_3&=\cos\phi\cos\theta, \end{aligned} \tag{2}$$

which satisfy $n_in_i=1$ identically. Because $\boldsymbol{E}_3$ is fixed in space, its body components evolve as

$$\dot{\boldsymbol{n}}=-\boldsymbol{\omega}\times\boldsymbol{n}, \tag{3}$$

where $\boldsymbol{\omega}$ is the angular velocity in body components. The contact point P is the point of the surface whose outward normal is $-\boldsymbol{n}$. Its position relative to G, $\boldsymbol{r}\equiv\boldsymbol{r}_{P/G}$, has components

$$r_i=-\frac{a_i^2n_i}{\sqrt{a_1^2n_1^2+a_2^2n_2^2+a_3^2n_3^2}}+b\,\delta_{i3}, \tag{4}$$

with no summation over i. We write $\mathring{\boldsymbol{r}}$ for the time derivative of these components; it follows from Eqs. (4) and (3). The Euler angles evolve as $(\dot\phi,\dot\theta,\dot\psi)^\top=\boldsymbol{\Phi}^{-1}\boldsymbol{\omega}$, where $\boldsymbol{\Phi}$ is the kinematic matrix of the 1–2–3 sequence. It is singular only at $\theta=\pm\pi/2$, far from the small rocking angles of interest, which is why this sequence is preferred over one that is singular at $\theta=0$.

B. Elimination of the contact force

Rolling without slipping, $\boldsymbol{v}_P=\boldsymbol{0}$, fixes the velocity and acceleration of G in terms of the rotation,

$$\boldsymbol{v}_G=-\boldsymbol{\omega}\times\boldsymbol{r}, \tag{5}$$

$$\boldsymbol{a}_G=-\dot{\boldsymbol{\omega}}\times\boldsymbol{r}-\boldsymbol{\omega}\times(\boldsymbol{\omega}\times\boldsymbol{r})-\boldsymbol{\omega}\times\mathring{\boldsymbol{r}}. \tag{6}$$

The balance of linear momentum, $\boldsymbol{f}_P-mg\boldsymbol{n}=m\boldsymbol{a}_G$, is substituted into the balance of angular momentum about G, $\boldsymbol{r}\times\boldsymbol{f}_P=\boldsymbol{I}\dot{\boldsymbol{\omega}}+\boldsymbol{\omega}\times\boldsymbol{I}\boldsymbol{\omega}$, where $\boldsymbol{I}$ is the inertia tensor about G. The contact force $\boldsymbol{f}_P$ then drops out, and [6, 11]

$$\big(\boldsymbol{I}-m\boldsymbol{X}^2\big)\dot{\boldsymbol{\omega}}=mg\,\boldsymbol{X}\boldsymbol{n}-m\boldsymbol{X}\boldsymbol{W}^2\boldsymbol{r}-m\boldsymbol{X}\boldsymbol{W}\mathring{\boldsymbol{r}}-\boldsymbol{W}\boldsymbol{I}\boldsymbol{\omega}, \tag{7}$$

where $\boldsymbol{X}$ and $\boldsymbol{W}$ are the skew-symmetric matrices of $\boldsymbol{r}$ and $\boldsymbol{\omega}$. Together with the Euler-angle kinematics, Eq. (7) is a system of six first-order equations.

Two features of Eq. (7) matter for what follows. First, the added inertia $-m\boldsymbol{X}^2$ and the terms $\boldsymbol{X}\boldsymbol{W}^2\boldsymbol{r}$ and $\boldsymbol{X}\boldsymbol{W}\mathring{\boldsymbol{r}}$ arise from eliminating $\boldsymbol{a}_G$ through the constraint; they are not separate physical effects. Second, the contact force is never computed, so the model gives no direct access to the normal reaction or to the tangential force required to sustain rolling.

III. SLIPPING-CONTACT FORMULATION

A. Releasing the rolling constraint

A Coulomb friction force has the form $\boldsymbol{F}=-\mu N\,\boldsymbol{v}_P/|\boldsymbol{v}_P|$. Under rolling, $\boldsymbol{v}_P\equiv\boldsymbol{0}$: the expression is undefined and, more fundamentally, there is no relative motion for friction to oppose, so a rolling body dissipates nothing at the contact. Moffatt and Shimomura make the same point for the spheroid [14]. In their analysis the slip velocity of the contact point reduces at leading order to a single horizontal component, and it is the frictional force opposing this slip that drives the secular rise; setting the slip to zero removes the mechanism.

Friction can therefore be introduced only after the constraint is released. Two things follow. The velocity of the center of mass, $\boldsymbol{v}\equiv\boldsymbol{v}_G$, is no longer determined by $\boldsymbol{\omega}$ and becomes an independent variable, which adds three states to give the nine-state system

$$\boldsymbol{y}=[\omega_1\ \omega_2\ \omega_3\ \phi\ \theta\ \psi\ v_1\ v_2\ v_3]^\top, \tag{8}$$

with $\boldsymbol{v}$ stored in body components. And the contact force can no longer be eliminated, so it has to be constructed explicitly. The balance of angular momentum returns to its plain form,

$$\boldsymbol{I}\dot{\boldsymbol{\omega}}+\boldsymbol{\omega}\times\boldsymbol{I}\boldsymbol{\omega}=\boldsymbol{r}\times\boldsymbol{f}_P+\boldsymbol{T}, \tag{9}$$

in which the terms specific to rolling in Eq. (7) no longer appear and $\boldsymbol{T}$ is an optional torque about the contact normal, used below for drilling friction.

B. Decomposition of the contact force

We write $\boldsymbol{f}_P = N\boldsymbol{n} + \boldsymbol{F}$ with $\boldsymbol{F}\cdot\boldsymbol{n} = 0$, where N is the normal reaction and $\boldsymbol{F}$ the tangential friction force. The slip velocity of the material point at P and its tangential part are

$$\boldsymbol{v}_P = \boldsymbol{v} + \boldsymbol{\omega}\times\boldsymbol{r}, \tag{10}$$

$$\boldsymbol{v}_P^t = \boldsymbol{v}_P - (\boldsymbol{v}_P\cdot\boldsymbol{n})\,\boldsymbol{n}. \tag{11}$$

The normal component of $\boldsymbol{v}_P$ vanishes for an exact solution; the projection in Eq. (11) removes any residual arising from numerical drift.

C. Normal reaction from the contact condition

Moffatt and Shimomura take the normal reaction to be of order Mg [14]. This is a leading-order statement within their gyroscopic-balance approximation, in which fast oscillations are averaged out. A rattleback, however, reverses through exactly such oscillations, and Sec. V E shows its normal reaction varying by a factor of almost 30 in a single run.

The normal reaction is a constraint force: it takes whatever value keeps P on the plane. The height of G above the plane follows from the geometry alone,

$$z_G = h(\boldsymbol{n}) = \Big(\sum_i a_i^2 n_i^2\Big)^{1/2} - b\,n_3 = -\boldsymbol{r}\cdot\boldsymbol{n}. \tag{12}$$

Differentiating once and using $\mathring{\boldsymbol{r}}\cdot\boldsymbol{n} = 0$, which holds because $\mathring{\boldsymbol{r}}$ is tangent to the surface at P, gives $\dot z_G = -\boldsymbol{r}\cdot\dot{\boldsymbol{n}}$. A second differentiation with $\ddot{\boldsymbol{n}} = -\dot{\boldsymbol{\omega}}\times\boldsymbol{n} - \boldsymbol{\omega}\times\dot{\boldsymbol{n}}$ yields an expression linear in $\dot{\boldsymbol{\omega}}$,

$$\ddot z_G = c + \boldsymbol{d}\cdot\dot{\boldsymbol{\omega}}, \tag{13}$$

$$c = -\mathring{\boldsymbol{r}}\cdot\dot{\boldsymbol{n}} + \boldsymbol{r}\cdot(\boldsymbol{\omega}\times\dot{\boldsymbol{n}}), \qquad \boldsymbol{d} = \boldsymbol{n}\times\boldsymbol{r}. \tag{14}$$

Both coefficients involve only quantities that the rolling formulation already computes.

Because $\boldsymbol{F}$ is tangential, the vertical balance of linear momentum reads

$$N - mg = m\ddot z_G. \tag{15}$$

For Coulomb friction both $\boldsymbol{F}$ and the drilling torque are proportional to N, so substituting $\boldsymbol{f}_P$ into Eq. (9) makes $\dot{\boldsymbol{\omega}}$ affine in N,

$$\dot{\boldsymbol{\omega}} = \boldsymbol{A}_0 + N(\boldsymbol{B} + \boldsymbol{A}_1), \tag{16}$$

with

$$\begin{aligned} &\boldsymbol{A}_0 = -\boldsymbol{I}^{-1}(\boldsymbol{\omega}\times\boldsymbol{I}\boldsymbol{\omega}), \qquad \boldsymbol{B} = \boldsymbol{I}^{-1}(\boldsymbol{r}\times\boldsymbol{n}),\\ &\boldsymbol{A}_1 = \boldsymbol{I}^{-1}\big[\boldsymbol{r}\times\hat{\boldsymbol{F}} + \hat{\boldsymbol{T}}\big], \end{aligned} \tag{17}$$

where $\hat{\boldsymbol{F}} = \boldsymbol{F}/N$ and $\hat{\boldsymbol{T}} = \boldsymbol{T}/N$. Combining Eqs. (13)–(16) gives a single scalar equation for N, whose solution is

$$N = \frac{m(g + c + \boldsymbol{d}\cdot\boldsymbol{A}_0)}{1 - m\,\boldsymbol{d}\cdot\boldsymbol{B} - m\,\boldsymbol{d}\cdot\boldsymbol{A}_1}. \tag{18}$$

Its evaluation costs one division per right-hand-side call and involves neither iteration nor approximation. For a tangential law that does not scale with N, such as the viscous law below, the torque $\boldsymbol{r}\times\boldsymbol{F}$ is moved into $\boldsymbol{A}_0$ and $\boldsymbol{A}_1 = \boldsymbol{0}$.

Well-posedness. Because $\boldsymbol{B} = -\boldsymbol{I}^{-1}\boldsymbol{d}$ and $\boldsymbol{I}$ is positive definite, $-m\,\boldsymbol{d}\cdot\boldsymbol{B} = m\,\boldsymbol{d}^{\top}\boldsymbol{I}^{-1}\boldsymbol{d} \geq 0$. In the frictionless case the denominator of Eq. (18) is therefore at least 1 and the solution always exists. Friction adds the term $-m\,\boldsymbol{d}\cdot\boldsymbol{A}_1$, which could in principle make the denominator vanish at large μ. That situation is the rigid-body inconsistency of Coulomb friction associated with Painlevé [15], not a numerical failure. The denominator was monitored in every computation and remained positive.

D. Translational equation

With $\boldsymbol{v}$ in body components, the balance of linear momentum becomes

$$\mathring{\boldsymbol{v}} = \frac{\boldsymbol{f}_P}{m} - g\,\boldsymbol{n} - \boldsymbol{\omega}\times\boldsymbol{v}, \tag{19}$$

where the last term accounts for the rotation of the body frame. The Euler-angle kinematics are unchanged. Equations (9), (18), and (19), with the kinematics, close the nine-state system once $\boldsymbol{F}$ and $\boldsymbol{T}$ are specified.

E. Tangential contact laws

Regularized Coulomb friction. The first law of Ref. [14] is a Coulomb law of magnitude μMg directed against the slip. We implement it as

$$\boldsymbol{F} = -\mu N\,\frac{\boldsymbol{v}_P^t}{\big(|\boldsymbol{v}_P^t|^2 + \varepsilon^2\big)^{1/2}}, \tag{20}$$

where the velocity scale ε smooths the sign function so that the equations remain integrable through zero slip. The instantaneous N of Eq. (18) replaces Mg. In the gyroscopic-balance analysis this law yields their Eq. (15), $\tan\theta = -(a/b)\tan\mu q(t - t_0)$, a transition completed in the finite interval $\Delta t = \pi/2|\mu q|$ [14].

Viscous friction. The second law of Ref. [14] is linear in the slip velocity. We write it as

$$\boldsymbol{F} = -c_v\,\boldsymbol{v}_P^t \tag{21}$$

with a constant coefficient c_v. It requires no regularization and is independent of N, so it enters $\boldsymbol{A}_0$ and Eq. (18) applies with $\boldsymbol{A}_1 = \boldsymbol{0}$. In the gyroscopic-balance analysis it yields their Eq. (16), $\tan\theta = e^{-\mu q(t - t_0)}$, an asymptotic approach on the secular time scale $|\mu q|^{-1}$ [14]. The two laws therefore differ in how they behave as the slip vanishes: the Coulomb force keeps magnitude μN down to slip speeds of order ε, whereas the viscous force decreases in proportion to the slip.

Drilling friction. A rigid point contact transmits no torque about the normal. A real contact occupies a small patch, and when the patch rotates relative to the table the friction distributed over it produces a torque about $\boldsymbol{n}$ that opposes the spin component $\omega_n = \boldsymbol{\omega}\cdot\boldsymbol{n}$. This torque is absent from Ref. [14]. Garcia and Hubbard included lumped spinning and slipping torques in their rattleback equations on the grounds that the contact is an area rather than a point [8]. The coupling between sliding and pivoting friction was analyzed by Contensou [16], developed by Zhuravlev into the Coulomb–Contensou model [17], and applied to the celt with an elliptic contact patch by Kudra and Awrejcewicz [13]. We use a single regularized term,

$$\boldsymbol{T} = -\mu N \rho\, \frac{\rho\,\omega_n}{\left(\rho^2\omega_n^2 + |\boldsymbol{v}_P^t|^2 + \varepsilon^2\right)^{1/2}}\,\boldsymbol{n}, \tag{22}$$

where ρ is an effective lever arm. The denominator weakens the torque when sliding dominates, which reproduces qualitatively the saturation of the Coulomb–Contensou analysis. It is not the integral model or its Padé approximants [13, 17], and we use it as a sensitivity study.

The lever arm can be estimated from Hertzian contact [18]. For a fully sliding circular patch of radius a_H under a Hertzian pressure distribution, integrating the Coulomb traction gives a torque $(3\pi/16)\mu N a_H$, so $\rho = (3\pi/16)a_H$. At the lowest point of the ellipsoid the principal radii of curvature are $a_1^2/a_3 = 2.0\,\mathrm{m}$ and $a_2^2/a_3 = 0.045\,\mathrm{m}$, with geometric mean $R_e = 0.30\,\mathrm{m}$. With $a_H = (3NR_e/4E^*)^{1/3}$ and $N = mg$, the patch radius is $a_H \approx 0.27\,\mathrm{mm}$ for steel on steel ($E^* \approx 115\,\mathrm{GPa}$) and $a_H \approx 0.88\,\mathrm{mm}$ for a polymer body ($E \approx 3\,\mathrm{GPa}$) on glass, giving $\rho \approx 0.16$–$0.52\,\mathrm{mm}$. We take $\rho = 0.5\,\mathrm{mm}$ as the representative value and examine $0 \le \rho \le 2\,\mathrm{mm}$.

F. Consistency with the rolling limit

As $\mu \to \infty$ the slip velocity tends to zero and the slipping formulation must reproduce Eq. (7). This holds by construction, since Eq. (7) follows from Eq. (9) once the exact contact force of a rolling body is substituted. The limit is the strongest test available, because it exercises the geometry, the normal-reaction solution, and the friction law at once.

IV. NUMERICAL METHOD AND VERIFICATION

A. Parameters

The physical parameters (Table I) are those of Ref. [6]. The sign of I_{12} sets the preferred sense, which for these values is $\omega_3 > 0$; we refer to $\omega_0 = +5\,\mathrm{rad\,s^{-1}}$ as the preferred sense and $\omega_0 = -5\,\mathrm{rad\,s^{-1}}$ as the non-preferred sense.

B. Implementation

The slipping model integrates the state of Eq. (8). Every run starts from a rolling state, $\boldsymbol{v}(0) = -\boldsymbol{\omega}(0)\times\boldsymbol{r}(0)$, so that $\boldsymbol{v}_P(0) = \boldsymbol{0}$ and the slipping and rolling models share the same initial condition; the large-μ limit can then be compared directly with the rolling solution. The geometric quantities $\boldsymbol{n}$, $\dot{\boldsymbol{n}}$, $\boldsymbol{r}$, and $\mathring{\boldsymbol{r}}$ are computed by one routine shared by both models. The rolling equations, Eq. (7), are integrated alongside every slipping run as a reference. Their contact force is recovered after integration from $\boldsymbol{f}_P = m(\boldsymbol{a}_G + g\,\boldsymbol{n})$ with $\boldsymbol{a}_G$ from Eq. (6), which makes the normal reaction and the friction demand of the rolling solution observable.

TABLE I. Physical parameters, after Kane and Levinson [6]. The product of inertia carries the chirality; setting it to zero removes the reversal.

Quantity	Symbol	Value
Mass	m	$1\,\mathrm{kg}$
Semi-axes	a_1, a_2, a_3	$0.20,\ 0.03,\ 0.02\,\mathrm{m}$
Offset of O above G	b	$0.005\,\mathrm{m}$
Moments of inertia	I_{11}, I_{22}, I_{33}	$2.0,\ 16.0,\ 17.5\times10^{-4}\,\mathrm{kg\,m^2}$
Product of inertia	I_{12}	$-0.5\times10^{-4}\,\mathrm{kg\,m^2}$
Initial spin	ω_0	$\pm5\,\mathrm{rad\,s^{-1}}$
Initial angles	ϕ, θ, ψ	$10^{-6}\,\mathrm{rad}$

The regularized Coulomb law contributes Jacobian entries of order $\mu N/\varepsilon$, about $5\times10^4\,\mathrm{N\,s\,m^{-1}}$ at the values used here, and makes the slipping system stiff. It was therefore solved with the variable-order backward-differentiation solver `ode15s` of MATLAB [19], with relative and absolute tolerances of 10^{-8} and 10^{-10}. For the Coulomb case the reversal time, final spin, and maximum rocking angle were unchanged to four significant figures when the tolerances were varied from 10^{-6} and 10^{-8} to 10^{-10} and 10^{-12}. The non-stiff rolling model was solved with the explicit Runge–Kutta pair `ode45` at tolerances of 10^{-10} and 10^{-12}. Solutions were written on a uniform grid of 20 001 points over 8 s. The solvers choose their own internal steps, so the output grid does not affect accuracy; it gives about 250 samples per period of the 11 Hz rocking, and the contact diagnostics below are evaluated at every output point.

The numerical part of the MATLAB code was also run under GNU Octave 8.4 [20], with the same results. An independent Python implementation, using the LSODA integrator [21] of SciPy [22], was written to cross-check the results and to render the animations. The two implementations agree on reversal times to within 0.002 s and on maximum rocking angles to within 0.01°. In the animations the orientation is assembled as $\boldsymbol{Q} = \boldsymbol{R}_1(\phi)\boldsymbol{R}_2(\theta)\boldsymbol{R}_3(\psi)$, which reproduces $\boldsymbol{Q}^{\top}\boldsymbol{E}_3 = \boldsymbol{n}$ to machine precision, and the body is drawn at its computed height $h(\boldsymbol{n})$, so it rests on the plane by construction.

C. Verification

Four checks were evaluated in every run. (i) *Contact residual.* The quantity $g_v = \boldsymbol{v}\cdot\boldsymbol{n} + \boldsymbol{r}\cdot\dot{\boldsymbol{n}}$, the rate of change of the gap between P and the plane, vanishes when Eq. (18) is satisfied. Its maximum over all runs was below $2\times10^{-7}\,\mathrm{m\,s^{-1}}$,

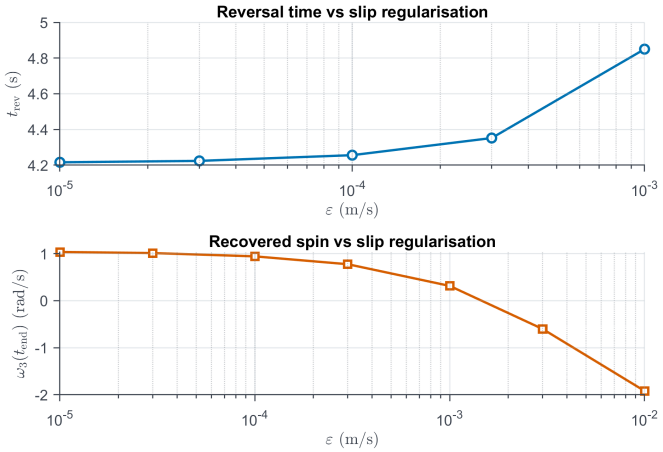


FIG. 2. Convergence of the Coulomb model ($\mu = 0.15$, non-preferred sense) in the regularization velocity ε. Top: reversal time, the first zero crossing of ω_3; points are missing where no reversal occurs within 8 s. Bottom: spin at $t = 8$ s.

TABLE II. Dependence of the Coulomb solution ($\mu = 0.15$, non-preferred sense) on the regularization velocity ε. A dash denotes no reversal within 8 s.

ε (m s^{-1})	10^{-2}	3×10^{-3}	10^{-3}	3×10^{-4}	10^{-4}	3×10^{-5}	10^{-5}
t_{rev} (s)	–	–	4.85	4.35	4.26	4.22	4.22
$\omega_3(8\,\mathrm{s})$	−1.92	−0.60	0.32	0.78	0.94	1.01	1.04

of the order of the solver tolerance. (ii) *Energy.* The total energy $E = \frac{1}{2}\boldsymbol{\omega}\cdot\boldsymbol{I}\boldsymbol{\omega} + \frac{1}{2}m|\boldsymbol{v}|^2 + mgh$ was conserved in the rolling model, with a drift below 10^{-10} J over 8 s, and decreased monotonically in every dissipative run. (iii) *Positive normal reaction.* $N > 0$ throughout, so the body never left the plane and the assumption of permanent contact is self-consistent. The denominator of Eq. (18) also remained positive. (iv) *Rolling limit.* As μ increases, the Coulomb solution converges to the rolling solution (Sec. V F); at $\mu = 1$ and 2 the reversal time agrees with the rolling value of 2.83 s to three significant figures.

D. Convergence in the regularization

The regularization requires care (Fig. 2, Table II). The reversal time and the final spin depend strongly on ε when it is comparable to the slip speeds, which are of order 10^{-2} m s^{-1} during reversal. At $\varepsilon = 10^{-2}$ and 3×10^{-3} m s^{-1} the smoothed law behaves like a viscous law at small slip and the body does not reverse within 8 s. At $\varepsilon = 10^{-3}$ m s^{-1}, a value that may appear small, the reversal occurs at 4.85 s, about 15% later than the converged value. Between $\varepsilon = 3\times10^{-5}$ and 10^{-5} m s^{-1} the reversal time changes by 0.007 s and the final spin by 0.02 rad s^{-1}. All Coulomb results below use $\varepsilon = 3\times10^{-5}$ m s^{-1}.

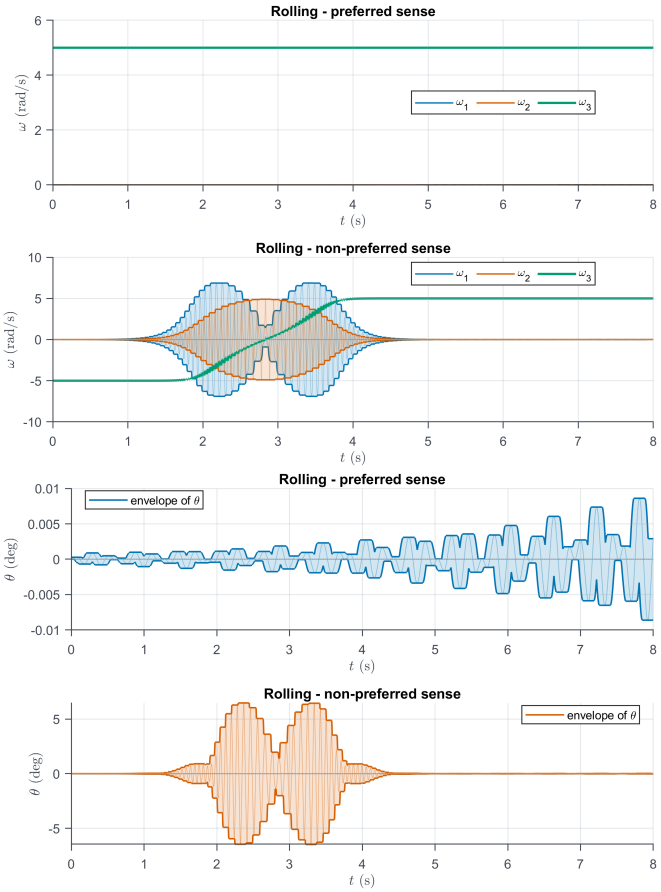


FIG. 3. Rolling solution. Upper pair: angular velocity components in the preferred and non-preferred senses. Lower pair: rocking angle θ. In the preferred sense ω_3 stays at $+5$ rad s^{-1} with no oscillation. In the non-preferred sense the spin reverses at $t = 2.83$ s, the energy passing through ω_1 and ω_2 oscillations of about ±7 and ±5 rad s^{-1} while θ reaches 6.5°. Because the model is conservative, ω_3 recovers its full magnitude.

V. RESULTS AND DISCUSSION

Table III collects the results for the four contact models in both senses of rotation. The subsections below discuss each model and then compare them.

A. Rolling solution

Figure 3 shows the rolling solution. In the preferred sense the spin remains at 5 rad s^{-1} and the rocking angle stays below 0.01°. In the non-preferred sense the rocking grows from $t \approx 1.3$ s, reaches $\theta_{\max} = 6.48°$, and ω_3 crosses zero at $t = 2.83$ s. The rocking then decays and the spin settles at $+5$ rad s^{-1}, its initial magnitude, as energy conservation requires. This reproduces the rolling formulation of Ref. [11] and agrees with Ref. [6].

TABLE III. Summary of results for $|\omega_0| = 5\,\mathrm{rad\,s^{-1}}$ over 8 s, with $\varepsilon = 3\times10^{-5}\,\mathrm{m\,s^{-1}}$. t_{rev} is the first zero crossing of ω_3, $\theta_{\max}$ the maximum rocking angle, and $\Delta E/E_0$ the fractional change of energy, with $E_0 = 0.169\,\mathrm{J}$. Energy was conserved or decreased monotonically in every run, and no contact law produced a reversal in the preferred sense.

Contact law	Sense	t_{rev} (s)	$\omega_3(8\,\mathrm{s})$ (rad s^{-1})	N/mg	$\theta_{\max}$	$\Delta E/E_0$
Rolling	preferred	–	+5.00	1.00	$< 0.01°$	0
Rolling	non-preferred	2.83	+5.00	0.20–5.86	6.48°	0
Coulomb, $\mu = 0.15$	preferred	–	+5.00	1.00	$< 0.01°$	0
Coulomb, $\mu = 0.15$	non-preferred	4.22	+1.01	0.76–1.30	0.84°	−12.4%
Coulomb + drilling, $\rho = 0.5\,\mathrm{mm}$	preferred	–	+1.64	1.00	$< 0.01°$	−11.5%
Coulomb + drilling, $\rho = 0.5\,\mathrm{mm}$	non-preferred	3.98	+0.02	0.76–1.30	0.84°	−12.9%
Viscous, $c_v = 500\,\mathrm{kg\,s^{-1}}$	preferred	–	+5.00	1.00	$< 0.01°$	0
Viscous, $c_v = 500\,\mathrm{kg\,s^{-1}}$	non-preferred	2.90	+1.40	0.40–2.63	3.09°	−11.9%

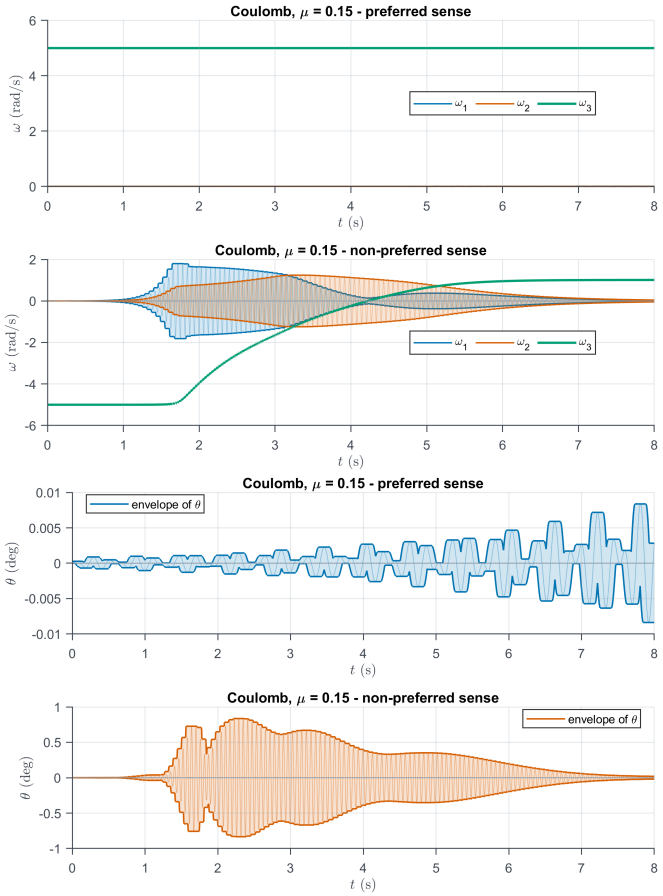


FIG. 4. Coulomb friction, $\mu = 0.15$. Upper pair: angular velocity components. Lower pair: rocking angle. The reversal occurs later than in rolling (4.22 s against 2.83 s), the rocking is limited to 0.84°, and the recovered spin is $+1.01\,\mathrm{rad\,s^{-1}}$ instead of $+5\,\mathrm{rad\,s^{-1}}$.

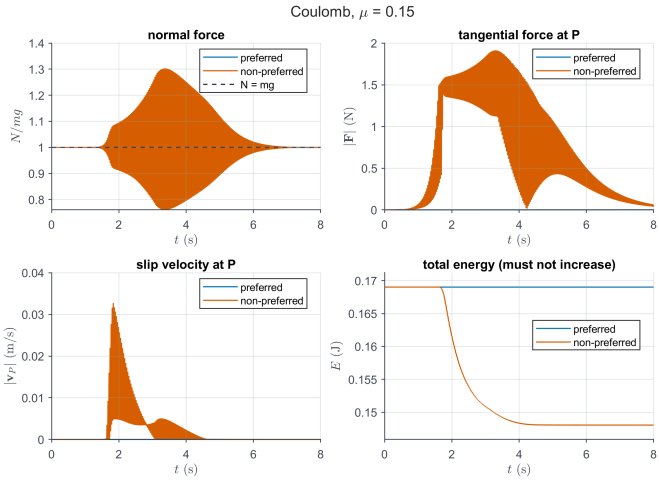


FIG. 5. Contact diagnostics for Coulomb friction, $\mu = 0.15$: normal reaction relative to mg, magnitude of the tangential force, slip speed of the contact point, and total energy. The preferred sense is flat in every panel. In the non-preferred sense N/mg ranges from 0.76 to 1.30, the slip reaches $32\,\mathrm{mm\,s^{-1}}$, and the energy falls monotonically by 12.4%.

B. Coulomb friction

With Coulomb friction at $\mu = 0.15$ (Figs. 4 and 5) the contact begins to slip as soon as the rocking grows, at speeds of up to $32\,\mathrm{mm\,s^{-1}}$. Friction limits the rocking to 0.84°, an eighth of the rolling value. Two features deserve comment. First, the reversal is delayed rather than accelerated, to 4.22 s from 2.83 s. Friction damps the very oscillation that carries the energy exchange, so the instability takes longer to complete. Second, the spin after reversal is a small fraction of the initial spin. About 12% of the initial energy is dissipated, almost all of it during the rocking phase, and the spin settles at $+1.01\,\mathrm{rad\,s^{-1}}$. A single reversal followed by a reduced spin agrees with the behavior of real rattlebacks, which reverse a finite number of times because of frictional losses [12], and differs qualitatively from the conservative rolling solution. In the preferred sense the Coulomb solution is indistinguishable from rolling: no rocking develops, the contact does not slip, and no energy is lost.

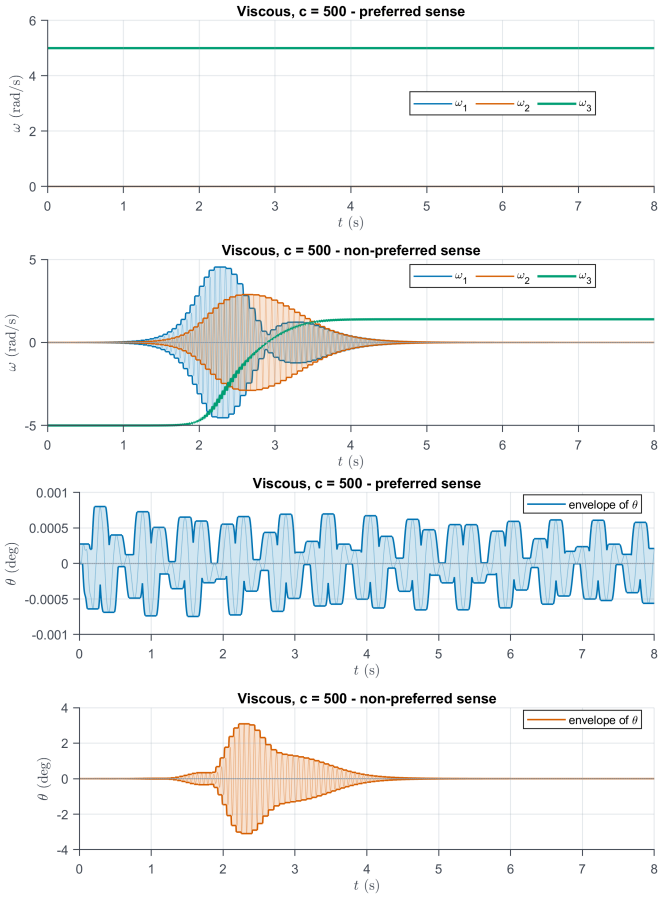


FIG. 6. Viscous friction, $c_v = 500\,\mathrm{kg\,s^{-1}}$. Upper pair: angular velocity components. Lower pair: rocking angle. The reversal occurs at 2.90 s, close to the rolling value, with partial recovery of the spin to $+1.40\,\mathrm{rad\,s^{-1}}$ and $\theta_{\max} = 3.09°$.

TABLE IV. Viscous coefficient sweep, non-preferred sense. A dash denotes no reversal within 8 s. The rolling reversal time is 2.83 s.

c_v (kg s^{-1})	1.5	10	50	200	500	2000
t_{rev} (s)	–	–	–	–	2.90	2.81
$\omega_3(8\,\mathrm{s})$	−5.00	−5.00	−4.78	−0.34	+1.40	+3.62

C. Viscous friction

With $c_v = 500\,\mathrm{kg\,s^{-1}}$ (Figs. 6 and 7) the viscous model reverses at 2.90 s, close to rolling, and retains more of its spin than the Coulomb model. The structural difference between the two laws is exposed by a sweep in c_v (Table IV). A coefficient of the same nominal size as the Coulomb case, $c_v = \mu mg/(1\,\mathrm{m\,s^{-1}}) \approx 1.5\,\mathrm{kg\,s^{-1}}$, produces a tangential force of order 10^{-2} N at the observed slip speeds. The contact is then effectively frictionless, the rocking modes are not excited, and the spin remains at $-5\,\mathrm{rad\,s^{-1}}$; the same holds for $c_v = 10\,\mathrm{kg\,s^{-1}}$. At $c_v = 50\,\mathrm{kg\,s^{-1}}$ the spin begins to decrease, and at $200\,\mathrm{kg\,s^{-1}}$ it reaches $-0.34\,\mathrm{rad\,s^{-1}}$ by $t = 8$ s without reversing. Reversal requires $c_v \gtrsim 500\,\mathrm{kg\,s^{-1}}$, and at

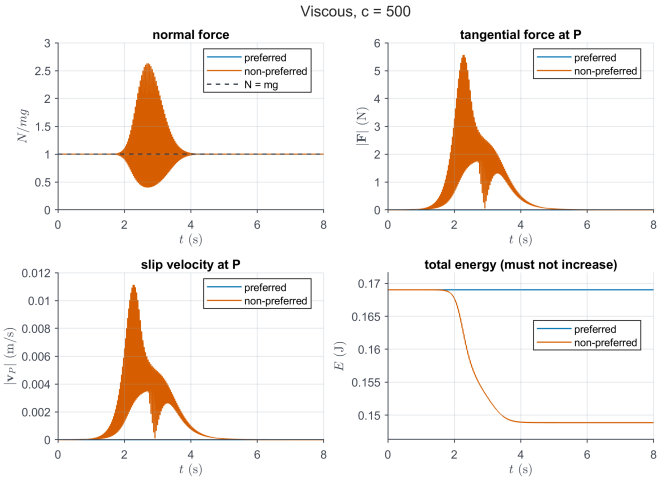


FIG. 7. Contact diagnostics for viscous friction, $c_v = 500\,\mathrm{kg\,s^{-1}}$. The normal reaction rises to $2.6\,mg$ and falls to $0.4\,mg$, a much larger excursion than with Coulomb friction because the rocking is less strongly damped.

TABLE V. Contact patch sweep for Coulomb friction with drilling torque, $\mu = 0.15$. $t_\times$ is the first zero crossing of ω_3 in the non-preferred sense; at $\rho = 2$ mm it is a spin-down rather than a reversal. The Hertzian estimate is $\rho \approx 0.16$–0.52 mm.

ρ (mm)	0	0.1	0.2	0.5	1.0	2.0
$t_\times$ (s)	4.22	4.19	4.15	3.98	3.68	2.99
$\omega_3(8\,\mathrm{s})$, pref.	5.00	4.33	3.66	1.64	0.00	0.00
$\omega_3(8\,\mathrm{s})$, non-pref.	1.01	0.78	0.52	0.02	0.02	0.00
$\theta_{\max}$ (deg)	0.84	0.84	0.84	0.84	0.72	0.06

$2000\,\mathrm{kg\,s^{-1}}$ the reversal time, 2.81 s, is close to the rolling value. In this regime the viscous law acts as a penalty that enforces approximate rolling.

The contrast follows from the property noted after Eq. (21). The instability needs a tangential force large enough to couple the rocking modes to the spin. A Coulomb contact supplies a force of order μN however small the slip, whereas a viscous contact supplies one only if c_v is large compared with N divided by a typical slip speed. In the gyroscopic-balance analysis of Ref. [14] the same distinction separates a transition completed in finite time [their Eq. (15)] from an exponential approach [their Eq. (16)]. In the rattleback it decides whether a reversal occurs at all.

D. Drilling friction

Drilling friction removes energy from the spin directly, without involving the rocking modes. At the representative lever arm $\rho = 0.5$ mm (Figs. 8 and 9) the rocking amplitude is unchanged at 0.84° and the reversal still occurs, slightly earlier, at 3.98 s, because the spin to be reversed has already been partly braked. The torque continues to act after the reversal, and the spin at $t = 8$ s is only $0.02\,\mathrm{rad\,s^{-1}}$. In the preferred

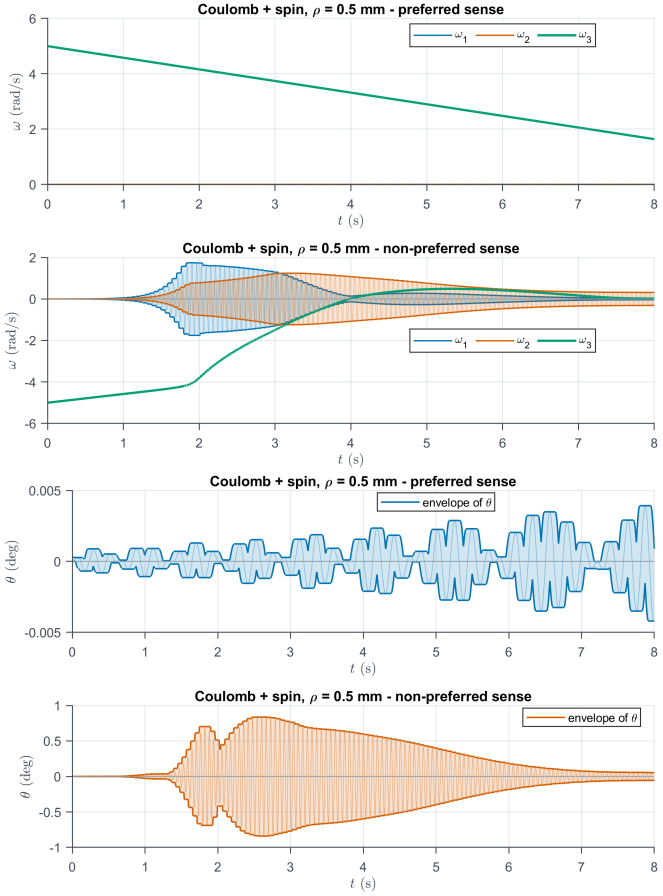


FIG. 8. Coulomb friction ($\mu = 0.15$) with a drilling torque, $\rho = 0.5$ mm. Upper pair: angular velocity components. Lower pair: rocking angle. In the non-preferred sense the reversal occurs at 3.98 s with the rocking unchanged at 0.84°, after which the drilling torque brings the spin nearly to rest. In the preferred sense the spin decays steadily from 5 to 1.64 rad s^{-1}.

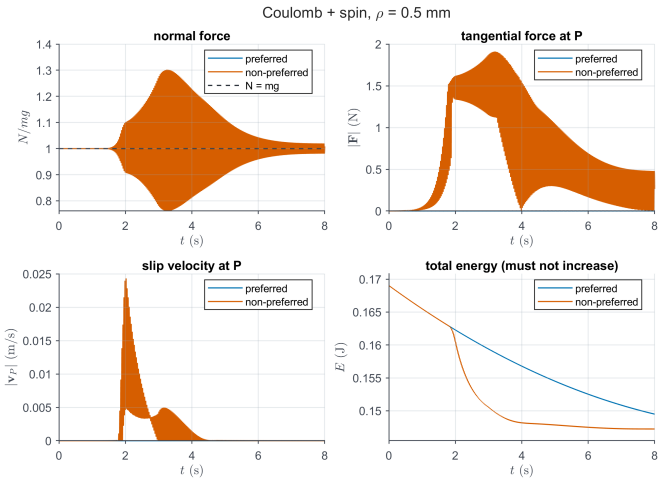


FIG. 9. Contact diagnostics for Coulomb friction with drilling torque, $\rho = 0.5$ mm. Unlike the other models, the energy decreases in the preferred sense as well, because the drilling torque acts on the spin directly.

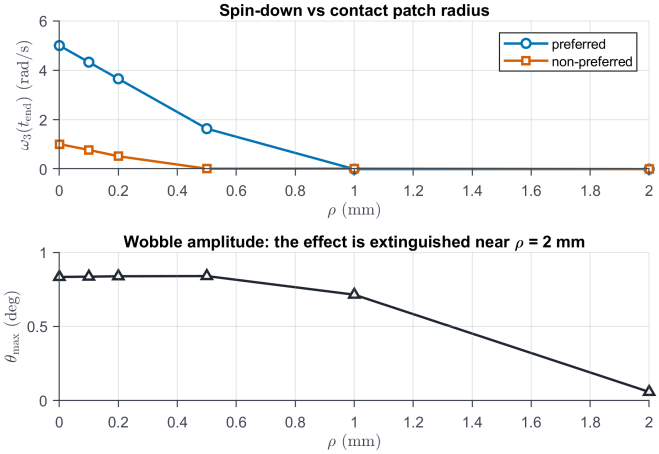


FIG. 10. Effect of the drilling lever arm ρ, Coulomb friction with $\mu = 0.15$. Top: spin at $t = 8$ s in both senses. Bottom: maximum rocking angle in the non-preferred sense.

sense, where no rocking develops, drilling friction is the only dissipation, and the spin decays from 5 to 1.64 rad s^{-1} in 8 s. A slow spin-down in the preferred sense is the expected behavior of a real celt and corrects a shortcoming of the conservative rolling model, in which the preferred spin persists indefinitely.

The sweep in ρ (Table V, Fig. 10) shows how sensitive the effect is to the assumed patch size. Within the Hertzian range the rocking survives essentially unchanged. For larger patches the drilling torque dominates. At $\rho = 2$ mm the torque $\mu N \rho \approx 2.9$ mN m, acting against $I_{33} = 1.75 \times 10^{-3}$ kg m^2, decelerates the spin at about 1.7 rad s^{-2} and brings 5 rad s^{-1} to rest in about 3 s. This happens before the instability can grow, and the maximum rocking angle falls to 0.06°. The zero crossing of ω_3 at 2.99 s is then a spin-down, not a reversal: the spin approaches zero and does not recover. This is a property of the model rather than a numerical artifact. A reversal requires the growth time of the instability to be shorter than the spin-down time set by drilling friction, and an oversized patch violates that condition.

E. The normal reaction

The bottom-left panel of Fig. 11 is the quantitative justification for solving Eq. (18). In the rolling solution, the exact no-slip motion, the normal reaction oscillates at the rocking frequency of about 11 Hz and spans $0.20\,mg$ to $5.86\,mg$, a factor of 29. The tangential force needed to prevent slip reaches $|\boldsymbol{F}|/N = 0.79$ near $t = 2.8$ s, so pure rolling requires a friction coefficient of at least about 0.8 for these parameters. Kondo and Nakanishi found a maximum ratio of about 0.2 in the cases they examined [10]; the ratio depends strongly on the rocking amplitude, which is large here. Even with Coulomb friction at $\mu = 0.15$, where the rocking is weak, the reaction departs from mg by up to 30%. The assumption $N = mg$ is therefore untenable during the rocking phase.

The consequence was quantified by repeating the Coulomb

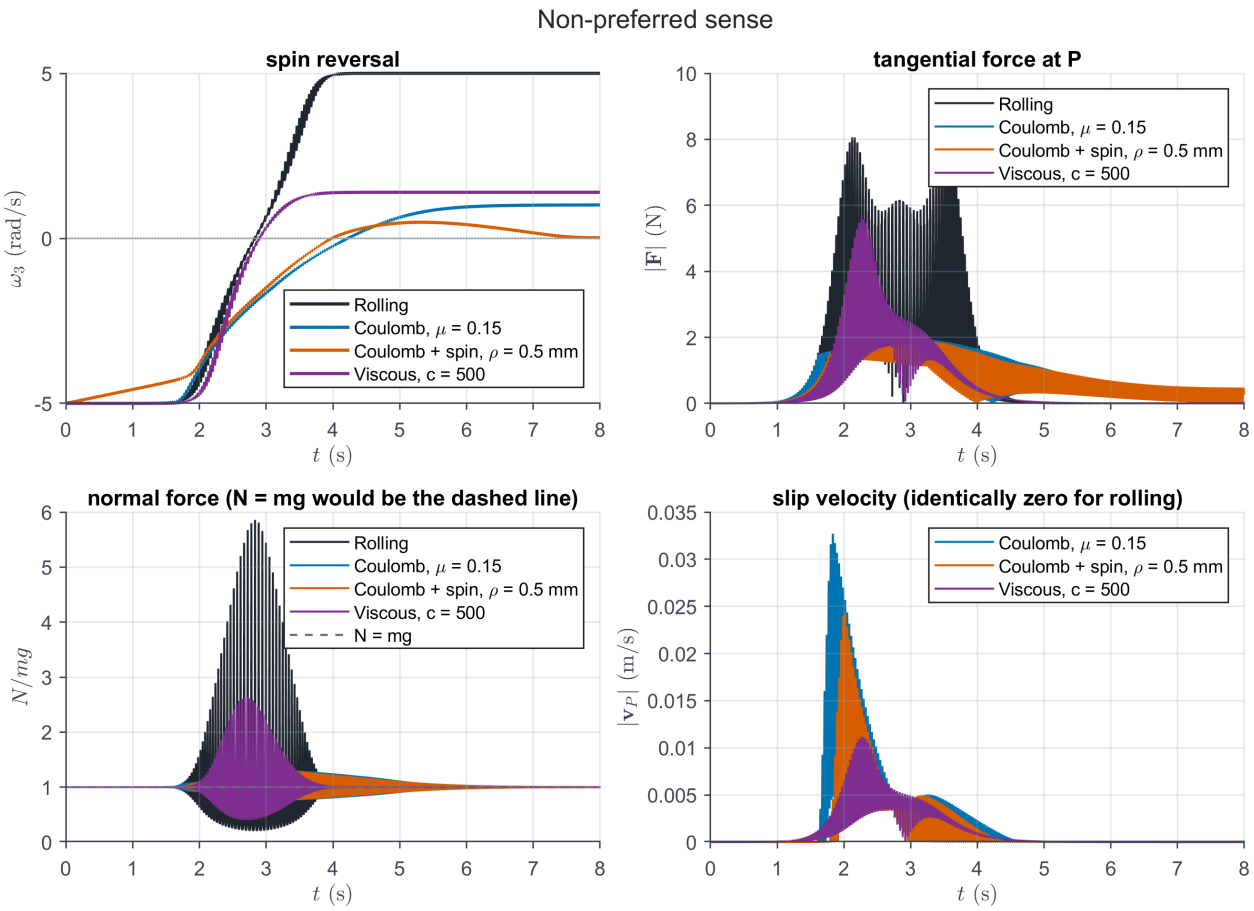


FIG. 11. Comparison of the four contact models in the non-preferred sense. Top left: spin ω_3. Top right: magnitude of the tangential force. Bottom left: normal reaction relative to mg; for the rolling solution it is the reaction required to sustain rolling and spans $0.20\,mg$ to $5.86\,mg$. Bottom right: slip speed, zero for rolling by definition.

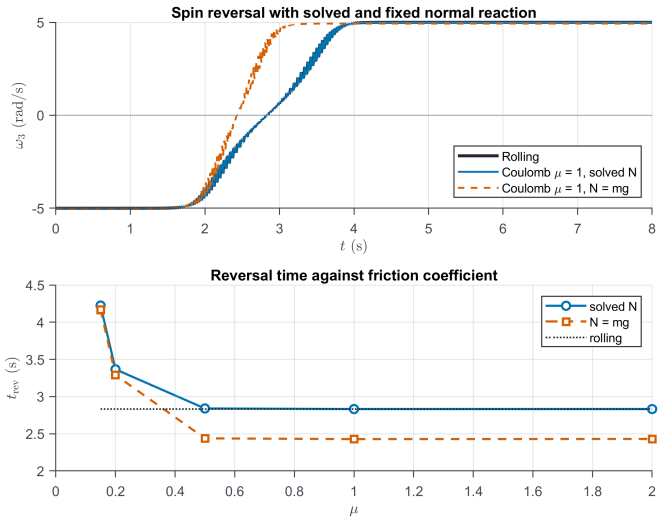


FIG. 12. Effect of approximating the normal reaction by mg. Top: ω_3 for Coulomb friction with $\mu = 1$, computed with N from Eq. (18) and with $N \equiv mg$, together with the rolling solution. Bottom: reversal time against μ; the dotted line is the rolling value.

TABLE VI. Reversal time (s) with the normal reaction from Eq. (18) and with $N \equiv mg$, Coulomb friction, non-preferred sense. The rolling value is 2.83 s.

μ	0.15	0.2	0.5	1.0	2.0
Solved N	4.22	3.37	2.84	2.83	2.83
$N \equiv mg$	4.17	3.29	2.44	2.43	2.43
Difference	−1.3%	−2.3%	−14%	−14%	−14%

runs with N held at mg in both the friction law and the torque balance (Fig. 12, Table VI). This simplification violates the contact condition, because it sets the vertical acceleration of G to zero while $h(\boldsymbol{n})$ continues to oscillate. In the weak-friction regime the error is modest, 1–2% at $\mu = 0.15$ and 0.2, because weak friction suppresses the rocking that makes N vary. As μ increases the error grows. With the solved reaction the reversal time converges to the rolling value, 2.84 s at $\mu = 0.5$ and 2.83 s at $\mu = 1$ and 2. With $N \equiv mg$ it approaches 2.43 s instead, an error of 14% that does not decrease with μ. Since convergence to the rolling limit is the strongest available test of the slipping formulation, the approximation also removes

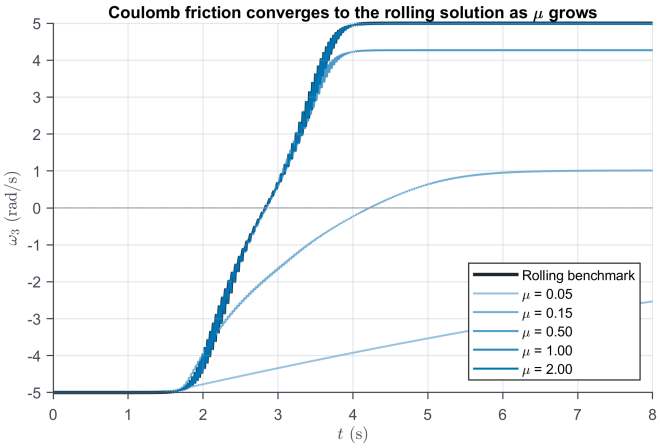


FIG. 13. Spin ω_3 under Coulomb friction for increasing μ, non-preferred sense, with the rolling solution (thick line).

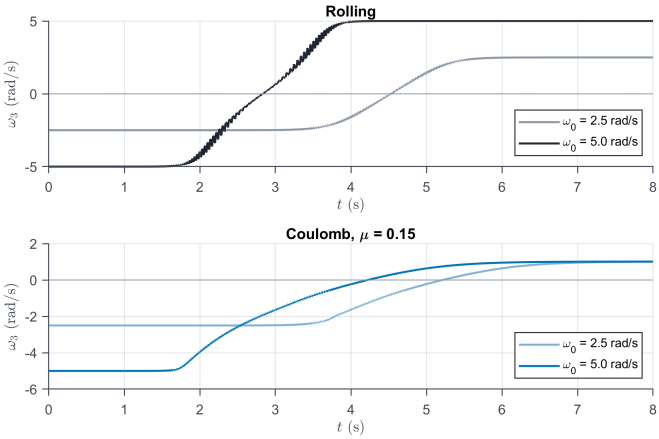


FIG. 14. Spin ω_3 for two initial spins in the non-preferred sense. Top: rolling. Bottom: Coulomb friction, $\mu = 0.15$.

the ability to verify the model.

F. Convergence to the rolling limit

Figure 13 shows the Coulomb solution for increasing friction coefficient. For $\mu = 0.05$ the friction force is too weak to couple the rocking to the spin: the spin has only fallen to $-2.53\,\mathrm{rad\,s^{-1}}$ by $t = 8\,\mathrm{s}$ and no reversal occurs. At $\mu = 0.15$ the reversal occurs at 4.22 s. For $\mu \geq 0.5$ the reversal time is 2.83–2.84 s, equal to the rolling value, and the recovered spin approaches the rolling value of $5\,\mathrm{rad\,s^{-1}}$, reaching $4.27\,\mathrm{rad\,s^{-1}}$ at $\mu = 0.5$ and $4.99\,\mathrm{rad\,s^{-1}}$ at $\mu = 1$. The normal reaction approaches the rolling range at the same time, with a maximum of $4.89\,mg$ at $\mu = 0.5$ and $5.76\,mg$ at $\mu = 1$, against $5.86\,mg$ for rolling. Once μ exceeds the friction demand of about 0.8 found in Sec. V E, the contact can sustain rolling throughout and the slipping solution follows the rolling one closely. This convergence is the rolling-limit check of Sec. IV C.

G. Dependence on the initial spin

Figure 14 compares two initial spins in the non-preferred sense. In the rolling model the smaller spin reverses later, at 4.53 s for $|\omega_0| = 2.5\,\mathrm{rad\,s^{-1}}$ against 2.83 s for $5\,\mathrm{rad\,s^{-1}}$, because the rocking instability grows more slowly at lower spin. Coulomb friction preserves this ordering, with reversals at 5.21 and 4.22 s. The final states differ more strikingly. The conservative rolling model returns the initial magnitude, 2.50 and $5.00\,\mathrm{rad\,s^{-1}}$, whereas under Coulomb friction both cases settle at about $1.01\,\mathrm{rad\,s^{-1}}$, almost independently of the initial spin. This suggests that, with friction, the spin left after reversal is set mainly by the dissipation during the rocking phase rather than by the initial condition.

H. Comparison of the angular velocities

Figure 15 compares the angular velocities of the four models. The ordering of the rocking amplitude in ω_1, rolling $>$ viscous $>$ Coulomb $\approx$ Coulomb with drilling, is the ordering of dissipation at the contact. All four models share the same underlying instability. They differ in how much of the energy released into the rocking is returned to the spin and how much is lost.

I. Animations

Figure 16 shows the four models at the same instant. The full animations (Appendix A) make two of the results directly visible: the rolling body rocks strongly and recovers its full spin, whereas the Coulomb body rocks weakly and comes almost to rest.

VI. LIMITATIONS

The model assumes a rigid, unilateral point contact that is never lost, with dissipation only through tangential slip and, optionally, drilling. Several effects are not included. The drilling torque, Eq. (22), uses a single effective lever arm rather than the integral Coulomb–Contensou model or its Padé approximants [13, 16, 17]. Rolling resistance, which arises from an asymmetric pressure distribution over a finite patch and shifts the line of action of the normal reaction, is neglected, as is contact compliance. If N were to become negative the body would lose contact and an impact model would be needed; this did not occur in any run reported here, and the code flags any such violation in its output. The friction coefficient is constant, and aerodynamic drag, small for dense bodies [8], is omitted. Finally, the Coulomb results depend on the regularization velocity unless it is well below the physical slip speeds (Sec. IV D). This is the largest numerical uncertainty, and converged values are quoted throughout.

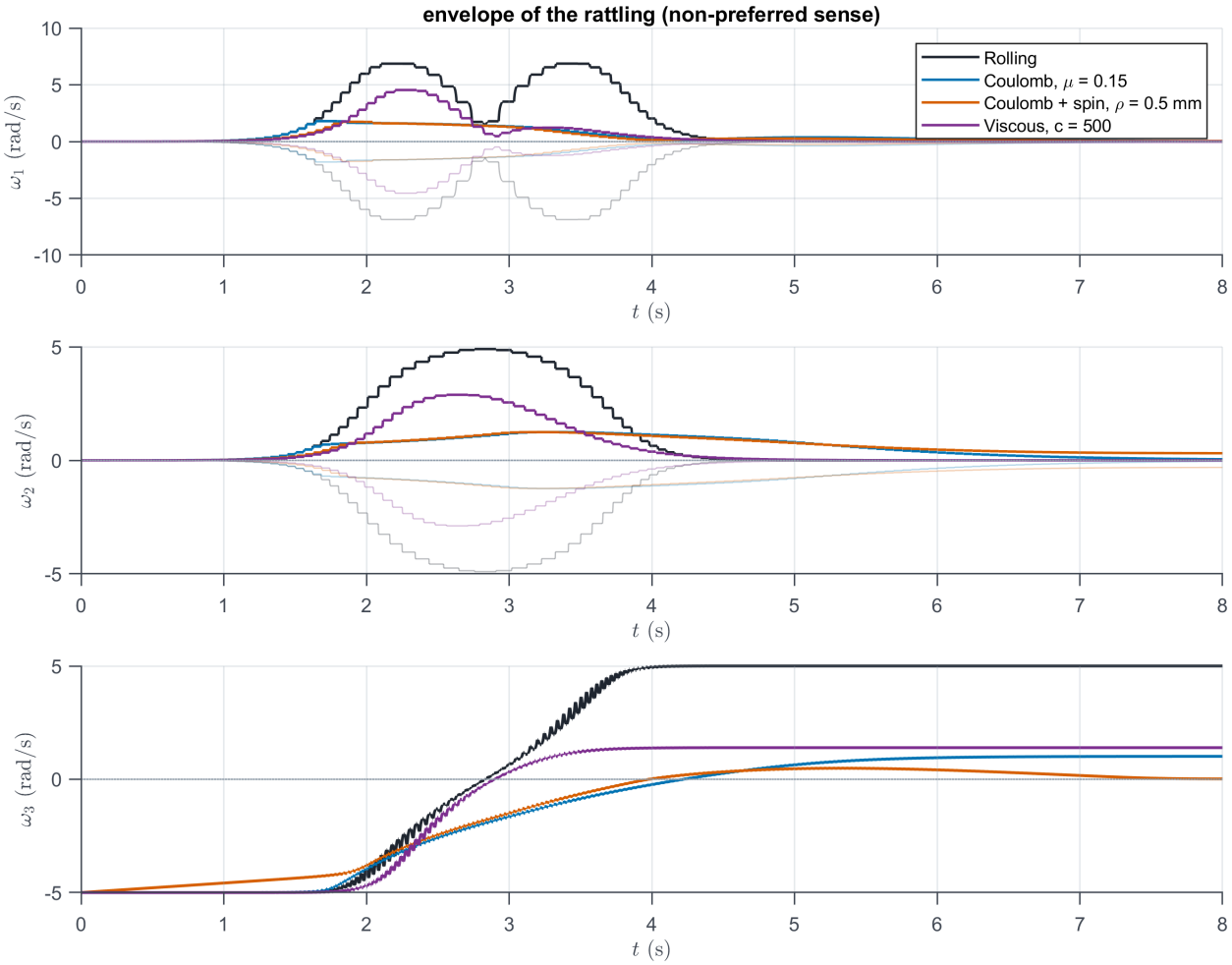


FIG. 15. Angular velocity components for the four contact models, non-preferred sense. For ω_1 and ω_2 the upper envelope of the rocking oscillation is drawn, with its mirror image faded. The models agree on ω_3 until the instability develops and then differ in the amplitude and duration of the rocking, which is where the friction laws remove energy.



FIG. 16. Frame at $t = 2.3$ s from the synchronized animation of the four contact models, non-preferred sense [23]. Each body is drawn at its true dimensions and at its computed height $h(\boldsymbol{n})$. The tilt is not exaggerated: the rolling body is near its 6.5° peak while the Coulomb body rocks by less than 1°.

VII. CONCLUSIONS

We have extended a rolling description of the rattleback to a contact that can slip and compared three tangential contact laws. The main findings are as follows.

(1) Friction cannot be added to a rolling model of the rattleback, because the slip velocity vanishes identically under rolling. The constraint has to be released, which raises the system from six to nine states and requires the contact force to be computed explicitly.

(2) The normal reaction must be solved from the contact condition rather than set equal to mg. Equation (18) gives it in closed form at the cost of one scalar division. In the rolling solution it varies between $0.20\,mg$ and $5.86\,mg$, and the approximation $N = mg$ produces a 14% error in the reversal time near the rolling limit while preventing convergence to that limit.

(3) The slipping formulation reproduces the rolling solution to three significant figures as μ grows, satisfies the contact condition to solver tolerance, and dissipates energy monoton-

ically for every friction law tested.

(4) Coulomb friction at $\mu = 0.15$ delays the reversal from 2.83 to 4.22 s, limits the rocking to 0.84°, and dissipates 12% of the initial energy, producing a single reversal to a reduced spin instead of the conservative behavior of the rolling model.

(5) The Coulomb and viscous laws differ structurally, not only quantitatively. Because the viscous force vanishes with the slip, a viscous coefficient of comparable nominal size produces no reversal at all, and the rolling behavior is recovered only as $c_v \to \infty$. This is the counterpart, for the rattleback, of the finite-time and asymptotic transitions obtained by Moffatt and Shimomura.

(6) Drilling friction is strongly sensitive to the contact patch. With a Hertzian lever arm, $\rho \approx 0.5$ mm, the reversal survives and the body then spins down, in both senses, as a real celt does. At $\rho = 2$ mm the drilling torque stops the body before the instability can develop, and the rattleback effect is suppressed.

DATA AVAILABILITY

The MATLAB source code that produced the figures, the Python implementation used for the cross-check and the animations, and six animations are openly available as Supplemental Material [23].

Appendix A: Animations and code

The Supplemental Material [23] contains six animations, each for $|\omega_0| = 5\,\mathrm{rad\,s^{-1}}$ over the first 6 s of motion: rolling without slipping in the non-preferred sense; rolling in the preferred and non-preferred senses side by side; Coulomb friction with $\mu = 0.15$; Coulomb friction with drilling torque, $\rho = 0.5$ mm; viscous friction with $c_v = 500\,\mathrm{kg\,s^{-1}}$; and all four contact models synchronized in a 2×2 grid. Playback is at 0.4 times real time so that the 11 Hz rocking is resolved at 40 frames per second. Each single-model animation shows the body together with $\omega_i(t)$ and $\theta(t)$ traced as time advances, and no deflection is exaggerated.

The MATLAB program `rattleback_simulator_123H.m` integrates all four contact models for both senses of the initial spin, performs the sweeps in ε, ρ, μ, and ω_0 and the comparison with $N \equiv mg$, prints the verification checks, and exports every figure in this paper except Figs. 1 and 16.

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[23] See Supplemental Material at `https://drive.google.com/drive/folders/1519soVugNzQgcVM6h5eciOCxXONTNRjv` for six animations and the source code (MATLAB and Python).